

Abstract Algebra Lecture Notes (Part II)

Factor Groups, Isomorphism Theorems, Permutation Groups,
Commutators, and Group Actions

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Mathematics III

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1 Normal Subgroup Intersections and Conjugacy in Permutation Groups

1.1 Completion of the Correspondence Theorem for Group Homomorphisms

Let $(G, \circ)$ and $(G', \circ')$ be two groups, and let $f : G \rightarrow G'$ be a group epimorphism (i.e., a surjective group homomorphism).

Let $\mathcal{S}(G)$ denote the set of all subgroups of G containing $\ker f$:

$$\mathcal{S}(G) = \{H \leq G \mid \ker f \subseteq H\},$$

and let $\mathcal{S}(G')$ denote the set of all subgroups of G' :

$$\mathcal{S}(G') = \{K \leq G'\}.$$

Define the mapping $\phi : \mathcal{S}(G) \rightarrow \mathcal{S}(G')$ by:

$$\phi(H) = f(H) \quad \text{for every } H \in \mathcal{S}(G).$$

Theorem 1.1 (Surjectivity of the Correspondence Map). *The mapping ϕ is surjective (onto).*

Proof. Let $K \in \mathcal{S}(G')$, which means that K is a subgroup of G' ($K \leq G'$). We must show that there exists some $H \in \mathcal{S}(G)$ such that $\phi(H) = K$.

Consider the complete pre-image of K under f :

$$H = f^{-1}(K) = \{x \in G \mid f(x) \in K\}.$$

We verify the required properties step-by-step:

1. **$f^{-1}(K)$ is a subgroup of G :** Since $e' \in K$ (where e' is the identity of G'), $f(e) = e' \in K$, so $e \in f^{-1}(K) \neq \emptyset$. Let $x, y \in f^{-1}(K)$. Then $f(x) \in K$ and $f(y) \in K$. Since $K \leq G'$,

$$f(xy^{-1}) = f(x)(f(y))^{-1} \in K \implies xy^{-1} \in f^{-1}(K).$$

Thus, $f^{-1}(K) \leq G$.

2. **$\ker f$ is contained in $f^{-1}(K)$:** Let $x \in \ker f$. Then $f(x) = e'$. Since $K \leq G'$, the identity element e' belongs to K . Thus:

$$f(x) = e' \in K \implies x \in f^{-1}(K).$$

Therefore:

$$\ker f \subseteq f^{-1}(K).$$

Consequently, $f^{-1}(K) \in \mathcal{S}(G)$.

3. **Image under ϕ :** Applying ϕ to $f^{-1}(K)$:

$$\phi(f^{-1}(K)) = f(f^{-1}(K)).$$

Since $f : G \rightarrow G'$ is given to be surjective (onto), for every element $k \in K \subseteq G'$, there exists $g \in G$ such that $f(g) = k$. By definition, $g \in f^{-1}(K)$, so $f(f^{-1}(K)) = K$. Thus:

$$\phi(f^{-1}(K)) = K.$$

Hence, ϕ is onto. Since ϕ was previously established to be injective, ϕ is a bijection. $\square$

1.2 Intersection of a Subgroup and a Normal Subgroup

Theorem 1.2. *Let G be a group, and let $H, K \leq G$. If K is a normal subgroup of G ($K \trianglelefteq G$), then:*

$$H \cap K \trianglelefteq H.$$

Proof. We proceed in two explicit steps:

1. **$H \cap K$ is a subgroup of H :** Since $H \leq G$ and $K \leq G$, the intersection of two subgroups is always a subgroup of G . Furthermore, $H \cap K \subseteq H$, so clearly:

$$H \cap K \leq H.$$

2. **$H \cap K$ is invariant under conjugation by elements of H :** Let $t \in H \cap K$ and let $h \in H$. By definition of set intersection:

$$t \in H \quad \text{and} \quad t \in K.$$

- Since $t \in H$ and $h \in H$, and H is a subgroup of G , by closure under multiplication and inverses:

$$hth^{-1} \in H.$$

- Since $h \in H \subseteq G$ and $t \in K$, and K is given to be a normal subgroup of G ($K \trianglelefteq G$), every conjugate of an element of K by any element of G remains in K :

$$hth^{-1} \in K \quad (\text{as } K \trianglelefteq G).$$

Combining both deductions:

$$hth^{-1} \in H \quad \text{and} \quad hth^{-1} \in K \implies hth^{-1} \in H \cap K.$$

Since for all $h \in H$ and all $t \in H \cap K$ we have $hth^{-1} \in H \cap K$, it follows by definition of normality that:

$$\therefore H \cap K \trianglelefteq H. \quad \square$$

1.3 Properties and Non-Transitivity of Normality

Note. 1. **Normality is NOT a transitive relation:** If $K \trianglelefteq H$ and $H \trianglelefteq G$, then K need not be a normal subgroup of G :

$$K \trianglelefteq H \text{ and } H \trianglelefteq G \not\Rightarrow K \trianglelefteq G.$$

(A concrete counterexample in S_4 is analyzed in Section 2.2).

2. **Intermediate normality:** If $H \trianglelefteq G$ and K is any intermediate subgroup of G containing H ($H \leq K \leq G$), then:

$$H \trianglelefteq K.$$

Proof. For all $k \in K$, since $K \subseteq G$, we have $k \in G$. Since $H \trianglelefteq G$, $kHk^{-1} = H$. Thus $kHk^{-1} = H$ holds for all $k \in K$, proving $H \trianglelefteq K$. $\square$

1.4 Conjugacy in the Symmetric Group S_n

Definition 1.1 (Conjugate Permutations). Two permutations p and $q \in S_n$ are said to be **conjugate** to each other in S_n if there exists a permutation $r \in S_n$ such that:

$$q = rpr^{-1}.$$

Definition 1.2 (Permutations of the Same Form / Cycle Type). Two permutations $p, q \in S_n$ are said to be of the **same form** (or have the same **cycle structure**) if:

1. Both p and q can be expressed as a product of the same number of disjoint cycles.
2. The corresponding disjoint cycles are of equal length.

Theorem 1.3. *Two permutations $p, q \in S_n$ are permutations of the same form if and only if they are conjugate in S_n .*

Example 1.3 (Finding the Conjugating Permutation in S_8). Let $p, q \in S_8$ be defined by:

$$\begin{aligned} p &= (1\ 2\ 3)(4\ 5\ 6)(7\ 8), \\ q &= (6\ 7)(2\ 3\ 4)(5\ 1\ 8). \end{aligned}$$

Observe that:

- p consists of two 3-cycles and one 2-cycle.
- q consists of two 3-cycles and one 2-cycle, since disjoint cycles commute:

$$q = (2\ 3\ 4)(5\ 1\ 8)(6\ 7).$$

Thus p and q have the exact same cycle form. By the theorem, they are conjugate. To find $r \in S_8$ such that $q = rpr^{-1}$, align the cycle representations of p and q directly:

$$\begin{aligned} p &= (1\ 2\ 3)(4\ 5\ 6)(7\ 8) \\ q &= (2\ 3\ 4)(5\ 1\ 8)(6\ 7) \end{aligned}$$

We define the permutation r by mapping each element of p to the corresponding element directly below it in q :

$$r = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 8 & 6 & 7 \end{pmatrix}.$$

Expressing r in disjoint cycle notation:

$$r = (1\ 2\ 3\ 4\ 5)(6\ 8\ 7).$$

Its inverse r^{-1} is obtained by reversing each cycle:

$$r^{-1} = (1\ 5\ 4\ 3\ 2)(6\ 7\ 8) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 1 & 2 & 3 & 4 & 7 & 8 & 6 \end{pmatrix}.$$

Verification of $rpr^{-1} = q$: Writing in two-row matrix form:

$$\begin{aligned}
 rpr^{-1} &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 8 & 6 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 5 & 6 & 4 & 8 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 1 & 2 & 3 & 4 & 7 & 8 & 6 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 8 & 6 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 2 & 3 & 1 & 5 & 8 & 7 & 4 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 3 & 4 & 2 & 1 & 7 & 6 & 5 \end{pmatrix} \\
 &= (185)(234)(67) \\
 &= (234)(518)(67) = q.
 \end{aligned}$$

Hence, $q = rpr^{-1}$ is verified.

2 Subgroups of S_4 , Inverses, and Failure of the Converse of Lagrange's Theorem

2.1 Explicit Structure of S_4 and the Alternating Group A_4

The symmetric group S_4 consists of all $4! = 24$ permutations on $\{1, 2, 3, 4\}$. Written out in full two-row and disjoint cycle forms:

$$S_4 = \left\{ I, \right. \\ (2\ 3\ 4), (2\ 4\ 3), (3\ 4), (2\ 3), (2\ 4), (1\ 2), (1\ 2)(3\ 4), \\ (1\ 2\ 3), (1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 2\ 4), (1\ 3\ 2), \\ (1\ 3\ 4\ 2), (1\ 3), (1\ 3\ 4), (1\ 3)(2\ 4), (1\ 3\ 2\ 4), \\ \left. (1\ 4\ 3\ 2), (1\ 4), (1\ 4\ 2), (1\ 4\ 3), (1\ 4\ 2\ 3), (1\ 4)(2\ 3) \right\}.$$

The alternating group A_4 is the subgroup of all even permutations in S_4 . Since $|S_4| = 24$, its order is:

$$|A_4| = \frac{|S_4|}{2} = \frac{24}{2} = 12.$$

Explicitly, A_4 contains the identity, the eight 3-cycles, and the three products of two disjoint transpositions:

$$A_4 = \left\{ I, \right. \\ (1\ 2\ 3), (1\ 3\ 2), (1\ 2\ 4), (1\ 4\ 2), (1\ 3\ 4), (1\ 4\ 3), (2\ 3\ 4), (2\ 4\ 3), \\ \left. (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3) \right\}.$$

Since $[S_4 : A_4] = 2$, it is immediate that:

$$A_4 \trianglelefteq S_4.$$

2.2 The Klein Four-Group and Non-Transitivity of Normality

The **Klein four-group** embedded in S_4 is:

$$V_4 = \{I, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}.$$

Notice that:

- $V_4 \subseteq A_4$.
- V_4 contains all permutations of cycle structure $(2, 2)$ in S_4 along with the identity.
- Since conjugation preserves cycle structure, for any $g \in S_4$ and any $v \in V_4$, gvg^{-1} must also have cycle structure $(2, 2)$ or be the identity, hence $gvg^{-1} \in V_4$.

Therefore:

$$V_4 \trianglelefteq S_4 \quad \text{and} \quad V_4 \trianglelefteq A_4.$$

Example 2.1 (Counterexample: Normality is NOT Transitive). Consider the subgroup H of V_4 generated by a single double transposition:

$$H = \{I, (1\ 2)(3\ 4)\} \leq V_4.$$

1. The index of H in V_4 is:

$$[V_4 : H] = \frac{|V_4|}{|H|} = \frac{4}{2} = 2.$$

Since every subgroup of index 2 is normal:

$$H \trianglelefteq V_4.$$

2. We already have $V_4 \trianglelefteq S_4$.
3. However, H is **not** a normal subgroup of S_4 ($H \not\trianglelefteq S_4$).

Proof. Take $g = (1\ 3) \in S_4$. Then $g^{-1} = (1\ 3)$. Conjugating $(1\ 2)(3\ 4) \in H$ by g :

$$g(1\ 2)(3\ 4)g^{-1} = (1\ 3)(1\ 2)(3\ 4)(1\ 3) = (3\ 2)(1\ 4) = (1\ 4)(2\ 3).$$

Notice that $(1\ 4)(2\ 3) \notin H$. Thus $gHg^{-1} \not\subseteq H$. □

Therefore, $H \trianglelefteq V_4$ and $V_4 \trianglelefteq S_4$, but $H \not\trianglelefteq S_4$.

Note. In S_n , if $H \trianglelefteq S_n$ and $p \in H$, then all permutations in S_n that have the same cycle form as p must also belong to H (since normal subgroups are unions of complete conjugacy classes).

2.3 Elements of Order 2 in Finite Groups of Even Order

Theorem 2.1. *If G is a finite group of even order, then there exists an element $a \in G$ such that $a \neq e$ and $a^2 = e$ (i.e., $|a| = 2$).*

Proof. Let G be a finite group with $|G| = 2k$ for some positive integer k . Every element $x \in G$ has a unique inverse $x^{-1} \in G$.

An element x satisfies $x = x^{-1}$ if and only if $x^2 = e$. The identity element e always satisfies $e = e^{-1}$ (i.e., $e^2 = e$).

Now consider the set of non-identity elements:

$$G \setminus \{e\}.$$

The cardinality of this set is:

$$|G \setminus \{e\}| = |G| - 1 = 2k - 1,$$

which is an **odd** integer.

We can partition $G \setminus \{e\}$ into two disjoint subsets:

1. Elements that are not their own inverses:

$$A = \{x \in G \setminus \{e\} \mid x \neq x^{-1}\}.$$

These elements can be grouped into disjoint pairs $\{x, x^{-1}\}$ of size 2. Consequently, $|A|$ is an **even** integer.

2. Elements that are their own inverses:

$$B = \{x \in G \setminus \{e\} \mid x = x^{-1}\}.$$

Since $(G \setminus \{e\}) = A \cup B$ and $A \cap B = \emptyset$:

$$|B| = |G \setminus \{e\}| - |A| = (\text{odd}) - (\text{even}) = \text{odd}.$$

Since $|B|$ is an odd integer, $|B| \geq 1$. Therefore, there exists at least one element $a \in G \setminus \{e\}$ such that:

$$a = a^{-1} \implies a^2 = e.$$

This element a has order $|a| = 2$. □

2.4 Failure of the Converse of Lagrange's Theorem

Lagrange's Theorem asserts that if G is a finite group and $H \leq G$, then $|H|$ divides $|G|$ ($|H| \mid |G|$).

The **converse of Lagrange's Theorem** would state:

If G is a finite group and $n \mid |G|$, then G must have a subgroup of order n .

The converse is false in general.

Example 2.2 (A_4 Has No Subgroup of Order 6). Consider the alternating group A_4 . We have $|A_4| = 12$. The integer 6 divides 12 ($6 \mid 12$). We prove that A_4 has **no** subgroup of order 6.

Proof. Suppose, for the sake of contradiction, that H is a subgroup of A_4 of order 6 ($|H| = 6$).

1. The index of H in A_4 is:

$$[A_4 : H] = \frac{|A_4|}{|H|} = \frac{12}{6} = 2.$$

Since any subgroup of index 2 is normal:

$$H \trianglelefteq A_4.$$

2. Since $|H| = 6$ is an even integer, by the preceding theorem, there exists an element $p \in H$ such that:

$$p \neq I \quad \text{and} \quad p^2 = I \quad (\text{i.e., } |p| = 2).$$

3. Recall that the elements of order 2 in A_4 are precisely the three double transpositions:

$$(1\ 2)(3\ 4), \quad (1\ 3)(2\ 4), \quad (1\ 4)(2\ 3).$$

Without loss of generality, let $p = (1\ 2)(3\ 4) \in H$.

4. Since $H \trianglelefteq A_4$, H is closed under conjugation by every element of A_4 .

- Conjugating p by $(1\ 2\ 3) \in A_4$:

$$(1\ 4)(2\ 3) = (1\ 2\ 3)(1\ 2)(3\ 4)(1\ 2\ 3)^{-1} \in H.$$

- Conjugating p by $(1\ 2\ 4) \in A_4$:

$$(1\ 3)(2\ 4) = (1\ 2\ 4)(1\ 2)(3\ 4)(1\ 2\ 4)^{-1} \in H.$$

5. Since $H \leq A_4$, both $(1\ 4)(2\ 3)$ and $(1\ 3)(2\ 4)$ must also belong to H . Together with $I \in H$ and $p = (1\ 2)(3\ 4) \in H$, we have:

$$V_4 = \{I, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\} \subseteq H.$$

6. Since V_4 is a subgroup of A_4 contained in H , $V_4 \leq H$. By Lagrange's Theorem, the order of V_4 must divide the order of H :

$$|V_4| \mid |H| \implies 4 \mid 6,$$

which is an outright contradiction, since $4 \nmid 6$.

Therefore, our assumption that A_4 contains a subgroup of order 6 is false. There is no subgroup of A_4 of order 6. $\square$

3 Quotient Groups (Factor Groups) and Canonical Projection

3.1 Coset Multiplication and Well-Definedness

Let G be a group and let $H \leq G$. The set of all left cosets of H in G is denoted by:

$$G/H = \{aH \mid a \in G\}.$$

We define a candidate binary operation $*$: $G/H \times G/H \rightarrow G/H$ by:

$$*(aH, bH) = abH \quad \text{or} \quad aH * bH = abH.$$

Remark. The operation $*$ **need not be a well-defined mapping** if H is merely a subgroup of G and not normal!

Example 3.1 (Counterexample: Operation Not Well-Defined When $H \not\trianglelefteq G$). Consider the symmetric group $G = S_3$ and the subgroup $H = \{I, (12)\}$. Compute the left cosets:

$$\begin{aligned} (123)H &= \{(123)I, (123)(12)\} = \{(123), (13)\} = (13)H, \\ (23)H &= \{(23)I, (23)(12)\} = \{(23), (132)\}. \end{aligned}$$

Since $(123)H = (13)H$, for $*$ to be well-defined, the product with $(23)H$ must yield the exact same coset:

$$\text{Must have:} \quad (123)H * (23)H = (13)H * (23)H.$$

Let us compute each side explicitly using the formula $xH * yH = xyH$:

- Left side:

$$(123)H * (23)H = (123)(23)H = (12)H = H.$$

- Right side:

$$(13)H * (23)H = (13)(23)H = (132)H = \{(132), (23)\} \neq H.$$

Therefore:

$$(123)H * (23)H \neq (13)H * (23)H.$$

Thus, the operation $*$ depends on the choice of coset representative, and so is **not well-defined**.

Theorem 3.1 (Well-Definedness Criterion and Quotient Group). *Let G be a group and $H \trianglelefteq G$. Then the operation:*

$$*: G/H \times G/H \rightarrow G/H, \quad aH * bH = abH$$

*is a well-defined binary operation on G/H . Furthermore, $(G/H, *)$ is a group, known as the **quotient group** (or **factor group**) of G with respect to H .*

Proof. We divide the proof into two parts:

Part 1: Well-Definedness. Let $(aH, bH) = (a'H, b'H)$ in $G/H \times G/H$. This means:

$$aH = a'H \quad \text{and} \quad bH = b'H.$$

We must show that $abH = a'b'H$.

- $aH = a'H \implies a \in a'H \implies a = a'h_1$ for some $h_1 \in H$.
- $bH = b'H \implies b \in b'H \implies b = b'h_2$ for some $h_2 \in H$.

Now calculate the coset abH :

$$\begin{aligned} abH &= (a'h_1)(b'h_2)H \\ &= a'h_1b'(h_2H) && (\text{as } h_2 \in H \implies h_2H = H) \\ &= a'h_1b'H. \end{aligned}$$

Since $H \trianglelefteq G$, left and right cosets coincide, so $b'H = Hb'$. Thus, $h_1b' \in Hb' = b'H$, which implies that there exists $h_3 \in H$ such that:

$$h_1b' = b'h_3.$$

Substituting this back:

$$\begin{aligned} abH &= a'(h_1b')H \\ &= a'(b'h_3)H \\ &= a'b'(h_3H) && (\text{as } h_3 \in H \implies h_3H = H) \\ &= a'b'H. \end{aligned}$$

Hence:

$$abH = a'b'H \implies aH * bH = a'H * b'H.$$

So $*$ is indeed a well-defined map, i.e., a binary operation on G/H .

Part 2: Group Axioms for $(G/H, *)$.

1. **Associativity:** Let $aH, bH, cH \in G/H$.

$$\begin{aligned} (aH * bH) * cH &= (abH) * cH \\ &= (ab)cH \\ &= a(bc)H && (\text{associativity in } G) \\ &= aH * (bcH) \\ &= aH * (bH * cH). \end{aligned}$$

So $*$ is associative.

2. **Identity Element:** The coset $eH = H$ serves as the identity element in G/H , because for any $aH \in G/H$:

$$aH * eH = (ae)H = aH \quad \text{and} \quad eH * aH = (ea)H = aH.$$

Thus $eH = H$ is the identity element in G/H with respect to $*$.

3. **Inverses:** For any $aH \in G/H$, the coset $a^{-1}H \in G/H$ is its inverse, because:

$$aH * a^{-1}H = (aa^{-1})H = eH = H \quad \text{and} \quad a^{-1}H * aH = (a^{-1}a)H = eH = H.$$

Thus, $(aH)^{-1} = a^{-1}H$.

Hence, $(G/H, *)$ is a group, known as the quotient group of G with respect to H . □

Note. If G is an abelian group, then the quotient group G/H is also abelian.

Proof. For any $aH, bH \in G/H$:

$$aH * bH = abH = baH = bH * aH \quad (\text{since } ab = ba \text{ in } G). \quad \square$$

3.2 The Canonical (Natural) Projection

Theorem 3.2. Let G be a group and $H \trianglelefteq G$. Then the mapping:

$$\nu : G \rightarrow G/H, \quad \nu(x) = xH \quad \forall x \in G$$

is a surjective group homomorphism with $\ker \nu = H$.

Proof. 1. **ν is a group homomorphism:** Let $x, y \in G$. Then:

$$\nu(xy) = (xy)H = xH * yH = \nu(x) * \nu(y).$$

Thus $\nu(xy) = \nu(x)\nu(y)$, so ν is a group homomorphism.

2. **ν is surjective (onto):** Let $t \in G/H$. By definition of G/H , t is a left coset of the form:

$$t = aH = \nu(a) \quad \text{for some } a \in G.$$

So ν is surjective. Hence ν is a surjective group homomorphism.

3. **Computation of $\ker \nu$:** By definition of the kernel of a homomorphism:

$$\begin{aligned} \ker \nu &= \{x \in G \mid \nu(x) = e_{G/H}\} \\ &= \{x \in G \mid \nu(x) = H\} \\ &= \{x \in G \mid xH = H\} \\ &= \{x \in G \mid x \in H\} \\ &= H. \end{aligned}$$

Thus, $\ker \nu = H$. □

Remark. This homomorphism ν is known as the **canonical projection**, **natural homomorphism**, or **quotient homomorphism**.

3.3 The Correspondence Theorem for Quotient Groups

Note. Recall from set theory that if $f : X \rightarrow Y$ is a mapping and $A_1, A_2 \subseteq X$:

1. $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$.
2. $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$, where equality holds if f is injective (one-to-one).

Proposition 3.3 (Lattice / Correspondence for Quotient Groups). *Let G be a group and $H \trianglelefteq G$. Then:*

1. *Any subgroup of G/H is of the form K/H , where K is a subgroup of G containing H ($H \leq K \leq G$).*
2. $K_1/H = K_2/H \implies K_1 = K_2$, where $K_1, K_2 \leq G$ containing H .
3. $(K_1 \cap K_2)/H = K_1/H \cap K_2/H$, where $K_1, K_2 \leq G$ containing H .

Proof. Consider the quotient group homomorphism $\nu : G \rightarrow G/H$ defined by $\nu(x) = xH$ for all $x \in G$. Then ν is surjective and $\ker \nu = H$.

By the general Correspondence Theorem (Theorem 1.1), there exists a bijection:

$$\phi : \mathcal{S}(G) \rightarrow \mathcal{S}'(G/H), \quad \phi(K) = \nu(K) \quad \forall K \in \mathcal{S}(G),$$

where $\mathcal{S}(G) = \{K \leq G \mid \ker \nu = H \subseteq K\}$ and $\mathcal{S}'(G/H) = \{T \leq G/H\}$.

Observe that for any $K \in \mathcal{S}(G)$:

$$\nu(K) = \{\nu(x) \mid x \in K\} = \{xH \mid x \in K\} = K/H.$$

1. Let $T \leq G/H$. Then $T \in \mathcal{S}'(G/H)$. Since ϕ is onto, there exists $K \in \mathcal{S}(G)$ such that:

$$T = \phi(K) = \nu(K) = K/H.$$

Since $K \in \mathcal{S}(G)$, we have $\ker \nu \subseteq K \implies H \leq K$. Thus $T = K/H$ for some $K \leq G$ containing H .

2. Let $K_1/H = K_2/H$.

$$\implies \nu(K_1) = \nu(K_2) \implies \phi(K_1) = \phi(K_2).$$

Since ϕ is injective (one-to-one):

$$K_1 = K_2.$$

3. Since ϕ is a bijection, we have:

$$\begin{aligned} (K_1 \cap K_2)/H &= \nu(K_1 \cap K_2) \\ &= \phi(K_1 \cap K_2) \\ &= \phi(K_1) \cap \phi(K_2) \quad (\text{as } \phi \text{ is a bijection}) \\ &= \nu(K_1) \cap \nu(K_2) \\ &= K_1/H \cap K_2/H. \end{aligned}$$

Therefore, $\boxed{(K_1 \cap K_2)/H = K_1/H \cap K_2/H}$. □

Note. If G is a group and $H \trianglelefteq G$, then any normal subgroup of G/H is of the form K/H , where K is a normal subgroup of G containing H ($H \subseteq K \trianglelefteq G$).

4 The Three Isomorphism Theorems

4.1 The First Isomorphism Theorem

Theorem 4.1 (Fundamental Theorem of Group Homomorphisms / First Isomorphism Theorem). *If G and G' are two groups and $f : G \rightarrow G'$ is a group homomorphism, then:*

$$G/\ker f \cong f(G).$$

In particular, if f is surjective (onto), then:

$$G/\ker f \cong G'.$$

Proof. As $f : G \rightarrow G'$ is a group homomorphism, $\ker f \trianglelefteq G$. Therefore, the quotient group $G/\ker f$ exists. Also, the image $f(G) = \text{Im } f$ is a subgroup of G' .

Define the map $\phi : G/\ker f \rightarrow f(G)$ by:

$$\phi(x \ker f) = f(x) \quad \forall x \in G.$$

We prove that ϕ is an isomorphism in four explicit steps:

Step 1: ϕ is well-defined. Let $x \ker f = y \ker f$ for some $x, y \in G$.

$$\begin{aligned} x \ker f = y \ker f &\implies x^{-1}y \in \ker f \\ &\implies f(x^{-1}y) = e' \\ &\implies f(x^{-1})f(y) = e' && \text{(as } f \text{ is a homomorphism)} \\ &\implies (f(x))^{-1}f(y) = e' \\ &\implies f(x) = f(y) \\ &\implies \phi(x \ker f) = \phi(y \ker f). \end{aligned}$$

Hence, ϕ is well-defined.

Step 2: ϕ is injective (one-to-one). Reversing the logic of Step 1:

$$\begin{aligned} \phi(x \ker f) = \phi(y \ker f) &\implies f(x) = f(y) \\ &\implies (f(x))^{-1}f(y) = e' \\ &\implies f(x^{-1})f(y) = e' \\ &\implies f(x^{-1}y) = e' \\ &\implies x^{-1}y \in \ker f \\ &\implies x \ker f = y \ker f. \end{aligned}$$

Therefore, ϕ is a one-to-one map.

Step 3: ϕ is surjective (onto). Let $t \in f(G)$. By definition of the image $f(G)$, there exists $a \in G$ such that:

$$t = f(a).$$

Since $a \in G \implies a \ker f \in G/\ker f$, and:

$$\phi(a \ker f) = f(a) = t.$$

So ϕ is onto.

Step 4: ϕ is a group homomorphism. For any $x \ker f, y \ker f \in G/\ker f$:

$$\begin{aligned}\phi(x \ker f \cdot y \ker f) &= \phi(xy \ker f) \\ &= f(xy) \\ &= f(x)f(y) \quad (\text{as } f \text{ is a group homomorphism}) \\ &= \phi(x \ker f)\phi(y \ker f).\end{aligned}$$

So ϕ is a group homomorphism.

Combining Steps 1–4, ϕ is an isomorphism from $G/\ker f$ onto $f(G)$. Therefore:

$$\therefore G/\ker f \cong f(G).$$

If f is onto, then $f(G) = G'$, and hence:

$$G/\ker f \cong G'.$$

□

4.2 Applications of the First Isomorphism Theorem

Example 4.1 (Application 1: $\mathbb{R}/\mathbb{Z} \cong S^1$). Let $(\mathbb{R}, +)$ be the additive group of real numbers, and let:

$$S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$$

be the circle group under complex multiplication. Show that $\mathbb{R}/\mathbb{Z} \cong S^1$.

Proof. Define $f : (\mathbb{R}, +) \rightarrow (S^1, \cdot)$ by:

$$f(x) = e^{2\pi i x} \quad \forall x \in \mathbb{R}.$$

1. **f is a group homomorphism:** For any $x, y \in \mathbb{R}$:

$$f(x + y) = e^{2\pi i(x+y)} = e^{2\pi i x} \cdot e^{2\pi i y} = f(x) \cdot f(y).$$

So f is a group homomorphism.

2. **f is onto:** Let $z \in S^1$. Since $|z| = 1$, z can be written in polar form as:

$$z = e^{i\theta} \quad \text{for some } \theta \in \mathbb{R}.$$

Setting $x = \frac{\theta}{2\pi} \in \mathbb{R}$, we find:

$$f\left(\frac{\theta}{2\pi}\right) = e^{2\pi i\left(\frac{\theta}{2\pi}\right)} = e^{i\theta} = z.$$

So f is onto.

3. **Computation of $\ker f$:**

$$\begin{aligned}\ker f &= \{x \in \mathbb{R} \mid f(x) = 1\} \\ &= \{x \in \mathbb{R} \mid e^{2\pi i x} = 1\} \\ &= \{x \in \mathbb{R} \mid \cos(2\pi x) + i \sin(2\pi x) = 1\} \\ &= \{x \in \mathbb{R} \mid 2\pi x = 2k\pi, k \in \mathbb{Z}\} \\ &= \{x \in \mathbb{R} \mid x \in \mathbb{Z}\} = \mathbb{Z}.\end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$\mathbb{R}/\ker f \cong S^1 \implies \boxed{\mathbb{R}/\mathbb{Z} \cong S^1}. \quad \square$$

Example 4.2 (Application 2: $\mathbb{Q}/\mathbb{Z} \cong P$). Let $P = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$ be the group of all roots of unity under complex multiplication. Show that $\mathbb{Q}/\mathbb{Z} \cong P$.

Proof. Define $f : (\mathbb{Q}, +) \rightarrow (P, \cdot)$ by:

$$f(x) = e^{2\pi i x} \quad \forall x \in \mathbb{Q}.$$

1. **f is well-defined into P :** If $x \in \mathbb{Q}$, then $x = \frac{m}{n}$ for some $m, n \in \mathbb{Z}$ with $n \neq 0$. Then:

$$(e^{2\pi i x})^n = (e^{2\pi i \frac{m}{n}})^n = e^{2\pi i m} = 1.$$

Thus $f(x) \in P$, so f is a well-defined map.

2. **f is a group homomorphism:** For all $x, y \in \mathbb{Q}$:

$$f(x+y) = e^{2\pi i(x+y)} = e^{2\pi i x} e^{2\pi i y} = f(x)f(y).$$

3. **f is onto:** Let $z \in P$. Then $z^n = 1$ for some $n \in \mathbb{Z}^+$. By Euler's formula:

$$\cos(2\pi m) + i \sin(2\pi m) = 1 \quad \forall m \in \mathbb{Z} \implies z^n = e^{2\pi i m} \implies z = e^{2\pi i \frac{m}{n}}.$$

Since $\frac{m}{n} \in \mathbb{Q}$, we have:

$$z = f\left(\frac{m}{n}\right) \quad \text{with } \frac{m}{n} \in \mathbb{Q}.$$

Therefore, f is onto.

4. **Computation of $\ker f$:**

$$\begin{aligned} \ker f &= \{x \in \mathbb{Q} \mid f(x) = 1\} \\ &= \{x \in \mathbb{Q} \mid e^{2\pi i x} = 1\} \\ &= \{x \in \mathbb{Q} \mid x \in \mathbb{Z}\} = \mathbb{Z}. \end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$\mathbb{Q}/\ker f \cong P \implies \boxed{\mathbb{Q}/\mathbb{Z} \cong P}. \quad \square$$

Example 4.3 (Application 3: $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$). Show that $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Proof. Define $f : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}_n, +)$ by:

$$f(x) = \bar{x}.$$

1. **f is a group homomorphism:**

$$f(x_1 + x_2) = \overline{x_1 + x_2} = \bar{x}_1 + \bar{x}_2 = f(x_1) + f(x_2).$$

2. **f is clearly onto:** For every $\bar{r} \in \mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$, $f(r) = \bar{r}$ with $r \in \mathbb{Z}$.

3. **Computation of $\ker f$:**

$$\begin{aligned}\ker f &= \{x \in \mathbb{Z} \mid f(x) = \bar{0}\} \\ &= \{x \in \mathbb{Z} \mid \bar{x} = \bar{0}\} \\ &= \{x \in \mathbb{Z} \mid x = nk, k \in \mathbb{Z}\} \\ &= n\mathbb{Z}.\end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$\mathbb{Z}/\ker f \cong \mathbb{Z}_n \implies \boxed{\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n}.$$

□

Example 4.4 (Application 4: $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong (\mathbb{R}^*, \cdot)$). Recall that:

$$\begin{aligned}GL(n, \mathbb{R}) &= \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}, \\ SL(n, \mathbb{R}) &= \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\}.\end{aligned}$$

Show that $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong (\mathbb{R}^*, \cdot)$.

Proof. Define $f : GL(n, \mathbb{R}) \rightarrow (\mathbb{R}^*, \cdot)$ by:

$$f(A) = \det(A) \quad \forall A \in GL(n, \mathbb{R}).$$

1. **f is a group homomorphism:** For any $A, B \in GL(n, \mathbb{R})$:

$$f(AB) = \det(AB) = \det(A) \det(B) = f(A)f(B).$$

2. **f is onto:** For any $\alpha \in \mathbb{R}^*$, consider the diagonal matrix:

$$A = \begin{pmatrix} \alpha & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}_{n \times n} \in GL(n, \mathbb{R}).$$

Then $\det(A) = \alpha \cdot 1 \cdots 1 = \alpha$. Thus $f(A) = \alpha$, so f is onto.

3. **Computation of $\ker f$:**

$$\begin{aligned}\ker f &= \{A \in GL(n, \mathbb{R}) \mid f(A) = 1\} \\ &= \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\} \\ &= SL(n, \mathbb{R}).\end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$GL(n, \mathbb{R})/\ker f \cong (\mathbb{R}^*, \cdot) \implies \boxed{GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong (\mathbb{R}^*, \cdot)}.$$

□

4.3 The Second Isomorphism Theorem (Diamond Isomorphism Theorem)

Theorem 4.2 (Second Isomorphism Theorem). *Let H and K be subgroups of a group G such that $K \trianglelefteq G$. Then:*

$$\frac{H}{H \cap K} \cong \frac{HK}{K}.$$

Proof. We construct the isomorphism by applying the First Isomorphism Theorem to an appropriately defined map on H :

1. As $K \trianglelefteq G$, by Theorem 1.2 we have $H \cap K \trianglelefteq H$. Therefore, the quotient group $\frac{H}{H \cap K}$ exists.
2. Since $K \trianglelefteq G$, HK is a subgroup of G and $K \leq HK$. Since $K \trianglelefteq G$, $K \trianglelefteq HK$. Therefore, the quotient group $\frac{HK}{K}$ exists.
3. Define $f : H \rightarrow HK/K$ by:

$$f(h) = hK \quad \forall h \in H.$$

Since $H \subseteq HK$, $hK \in HK/K$, so f is a well-defined map.

4. **f is a group homomorphism:** For any $h_1, h_2 \in H$:

$$\begin{aligned} f(h_1 h_2) &= (h_1 h_2)K \\ &= (h_1 K)(h_2 K) \\ &= f(h_1) f(h_2). \end{aligned}$$

Thus f is a group homomorphism.

5. **f is onto:** Let $t \in HK/K$. Then $t = xK$ for some $x \in HK$. Since $x \in HK$, $x = hk$ for some $h \in H$ and $k \in K$. Then:

$$t = xK = (hk)K = h(kK) = hK = f(h).$$

Hence, f is onto.

6. **Computation of $\ker f$:**

$$\begin{aligned} \ker f &= \{h \in H \mid f(h) = K\} \\ &= \{h \in H \mid hK = K\} \\ &= \{h \in H \mid h \in K\} \\ &= H \cap K. \end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$H / \ker f \cong HK / K \implies \boxed{\frac{H}{H \cap K} \cong \frac{HK}{K}}.$$

□

4.4 The Third Isomorphism Theorem

Theorem 4.3 (Third Isomorphism Theorem). *Let H and K be normal subgroups of a group G such that $H \subseteq K$ ($H \trianglelefteq G, K \trianglelefteq G$, and $H \subseteq K$). Then:*

$$\frac{G/H}{K/H} \cong \frac{G}{K}.$$

Proof. 1. As $H, K \trianglelefteq G$, the quotient groups G/H and G/K exist. Also, $H \subseteq K$, so H is a normal subgroup of K . Therefore, the quotient group K/H exists. As $K \trianglelefteq G$ such that $H \subseteq K$, by the Correspondence Theorem $K/H \trianglelefteq G/H$. Therefore, the quotient group $\frac{G/H}{K/H}$ exists.

2. Define $f : G/H \rightarrow G/K$ by:

$$f(xH) = xK \quad \forall x \in G.$$

Well-definedness of f :

$$\begin{aligned} xH = yH &\implies x^{-1}y \in H \subseteq K \\ &\implies x^{-1}y \in K \\ &\implies xK = yK \\ &\implies f(xH) = f(yH). \end{aligned}$$

So f is a well-defined map.

3. **f is a group homomorphism:** For any $xH, yH \in G/H$:

$$\begin{aligned} f(xH \cdot yH) &= f(xyH) \\ &= xyK \\ &= xK \cdot yK \\ &= f(xH)f(yH). \end{aligned}$$

Thus f is a group homomorphism.

4. **f is onto:** Let $t \in G/K$. Then $t = xK$ for some $x \in G$. Since $x \in G \implies xH \in G/H$, and:

$$f(xH) = xK = t.$$

So f is onto.

5. **Computation of $\ker f$:**

$$\begin{aligned} \ker f &= \{xH \in G/H \mid f(xH) = K\} \\ &= \{xH \in G/H \mid xK = K\} \\ &= \{xH \in G/H \mid x \in K\} \\ &= K/H. \end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$(G/H)/\ker f \cong G/K \implies \boxed{\frac{G/H}{K/H} \cong \frac{G}{K}}.$$

□

4.5 The Isomorphism $S_n/A_n \cong (\{1, -1\}, \cdot)$

Problem 4.5. Show that $S_n/A_n \cong (\{1, -1\}, \cdot)$.

Proof. Define the signature map $\text{sgn} : S_n \rightarrow (\{1, -1\}, \cdot)$ by:

$$\text{sgn}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is an even permutation,} \\ -1 & \text{if } \sigma \text{ is an odd permutation.} \end{cases}$$

1. Since the parity of a product of permutations satisfies $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$, sgn is a group homomorphism.
2. Since $n \geq 2$, S_n contains transpositions (which are odd), so sgn is surjective onto $\{1, -1\}$.
3. The kernel is:

$$\ker(\text{sgn}) = \{\sigma \in S_n \mid \text{sgn}(\sigma) = 1\} = A_n.$$

By the First Isomorphism Theorem:

$$S_n / \ker(\text{sgn}) \cong \{1, -1\} \implies \boxed{S_n/A_n \cong (\{1, -1\}, \cdot)}.$$

□

5 Products of Subgroups and Order Formulas

5.1 Criterion for HK to be a Subgroup

Definition 5.1 (Product of Subgroups). Let H and K be two subgroups of a group G . The **product** of H and K is defined as the set:

$$HK = \{hk \mid h \in H, k \in K\}.$$

Theorem 5.1. Let H and K be subgroups of a group G . Then HK is a subgroup of G if and only if $HK = KH$:

$$HK \leq G \iff HK = KH.$$

Proof. We prove both implications:

($\implies$) **Assume $HK \leq G$. We show $HK = KH$.**

1. Let $x \in KH$. Then $x = kh$ for some $k \in K$ and $h \in H$. Consider its inverse:

$$x^{-1} = (kh)^{-1} = h^{-1}k^{-1}.$$

Since $h \in H \implies h^{-1} \in H$ and $k \in K \implies k^{-1} \in K$, we have $x^{-1} \in HK$. Since HK is given to be a subgroup of G , the inverse of any element in HK also belongs to HK :

$$(x^{-1})^{-1} \in HK \implies x \in HK.$$

Therefore:

$$KH \subseteq HK.$$

2. Conversely, let $y \in HK$. Since $HK \leq G$, $y^{-1} \in HK$. Thus $y^{-1} = hk$ for some $h \in H, k \in K$. Taking the inverse:

$$y = (y^{-1})^{-1} = (hk)^{-1} = k^{-1}h^{-1}.$$

Since $k^{-1} \in K$ and $h^{-1} \in H$, $y \in KH$. Therefore:

$$HK \subseteq KH.$$

Combining both inclusions:

$$HK = KH.$$

($\impliedby$) **Assume $HK = KH$. We show $HK \leq G$.**

1. Since $e \in H$ and $e \in K$, $e = ee \in HK \neq \emptyset$.
2. **Closure under multiplication:** Let $x, y \in HK$. Then $x = h_1k_1$ and $y = h_2k_2$ for $h_1, h_2 \in H$ and $k_1, k_2 \in K$.

$$\begin{aligned} xy &= (h_1k_1)(h_2k_2) \\ &= h_1(k_1h_2)k_2. \end{aligned}$$

Since $k_1h_2 \in KH$ and $KH = HK$, there exist $h_3 \in H$ and $k_3 \in K$ such that $k_1h_2 = h_3k_3$. Substituting:

$$xy = h_1(h_3k_3)k_2 = (h_1h_3)(k_3k_2).$$

Since $H \leq G \implies h_1h_3 \in H$, and $K \leq G \implies k_3k_2 \in K$, it follows that:

$$xy \in HK.$$

3. Inverses: Let $x = hk \in HK$. Then:

$$x^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in KH.$$

Since $KH = HK$, $x^{-1} \in HK$.

Therefore, $HK \leq G$. □

Theorem 5.2. Let $H, K \leq G$. If $K \trianglelefteq G$, then $HK = KH$, and consequently $HK \leq G$.

Proof. Let $x \in HK \implies x = hk$ for some $h \in H, k \in K$. Notice that:

$$x = hk = (hkh^{-1})h.$$

Since $K \trianglelefteq G$, $hkh^{-1} \in K$. Let $k' = hkh^{-1} \in K$. Then:

$$x = k'h \in KH \implies HK \subseteq KH.$$

Similarly, let $y \in KH \implies y = kh$ for some $k \in K, h \in H$. Notice that:

$$y = kh = h(h^{-1}kh).$$

Since $K \trianglelefteq G$, $h^{-1}kh \in K$. Let $k'' = h^{-1}kh \in K$. Then:

$$y = hk'' \in HK \implies KH \subseteq HK.$$

Hence $HK = KH$. By the preceding theorem, $HK \leq G$. □

Note. If $H, K \leq G$ such that **either** $H \trianglelefteq G$ or $K \trianglelefteq G$, then $HK \leq G$ and $KH \leq G$.

5.2 Formula for the Order of HK

Theorem 5.3. If H and K are finite subgroups of a group G , then:

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof. Notice that the product set HK can be written as the union of left cosets of K :

$$HK = \bigcup_{h \in H} hK.$$

We determine how many distinct cosets of K appear in this union. For any $h_1, h_2 \in H$:

$$\begin{aligned} h_1K = h_2K &\iff h_1^{-1}h_2 \in K \\ &\iff h_1^{-1}h_2 \in H \cap K && (\text{since } h_1, h_2 \in H \implies h_1^{-1}h_2 \in H) \\ &\iff h_1(H \cap K) = h_2(H \cap K). \end{aligned}$$

Therefore, the number of distinct left cosets of K of the form hK ($h \in H$) is equal to the number of distinct left cosets of $H \cap K$ in H . This number is precisely the index:

$$[H : H \cap K] = \frac{|H|}{|H \cap K|}.$$

Each left coset hK contains exactly $|K|$ elements, and distinct cosets are completely disjoint. Therefore, the total number of elements in HK is:

$$|HK| = [H : H \cap K] \cdot |K| = \frac{|H|}{|H \cap K|} \cdot |K| = \frac{|H||K|}{|H \cap K|}. \quad \square$$

5.3 Comprehensive Practice Problems on Subgroup Orders

Proposition 5.4. *If $f : G \rightarrow G'$ is a group homomorphism and $a \in G$ is an element of finite order $|a| = n$, then $|f(a)|$ is finite and divides $|a|$:*

$$|f(a)| \mid |a|.$$

Furthermore, if f is injective (one-to-one), then $|f(a)| = |a|$.

Proof. Let $|a| = n \implies a^n = e$. Applying f :

$$\begin{aligned} f(a^n) &= f(e) \\ \implies \underbrace{f(a)f(a) \cdots f(a)}_{n \text{ times}} &= e' \\ \implies (f(a))^n &= e'. \end{aligned}$$

Since $(f(a))^n = e'$, the order $|f(a)|$ is finite and must divide n :

$$|f(a)| \mid n \implies |f(a)| \mid |a|.$$

If f is one-to-one: let $m = |f(a)|$. Then $(f(a))^m = e' \implies f(a^m) = f(e) \implies a^m = e \implies n \mid m$. Since $m \mid n$ and $n \mid m$, $m = n$, so $|f(a)| = |a|$. $\square$

Problem 5.2. Are $\mathbb{Z}_4$ and V_4 isomorphic?

Solution. **No.** $\mathbb{Z}_4$ is a cyclic group and contains an element of order 4 (namely $\bar{1}$ and $\bar{3}$, since $|\bar{1}| = 4$). However, V_4 contains no element of order 4 (every non-identity element in V_4 has order 2). Since an isomorphism must preserve element orders, $\mathbb{Z}_4 \not\cong V_4$. $\square$

Problem 5.3. Show that if $H \leq S_n$, then either all permutations in H are even, or exactly half of them are even.

Solution. Consider the intersection $H \cap A_n$. By Theorem 1.2, since $A_n \trianglelefteq S_n$, we have $H \cap A_n \trianglelefteq H$. By the Second Isomorphism Theorem:

$$\frac{H}{H \cap A_n} \cong \frac{HA_n}{A_n}.$$

Since $A_n \leq HA_n \leq S_n$ and $[S_n : A_n] = 2$, there are only two possibilities for the subgroup HA_n :

1. $HA_n = A_n \implies H \subseteq A_n$. In this case, every permutation in H is even.
2. $HA_n = S_n$. In this case:

$$[H : H \cap A_n] = [S_n : A_n] = 2.$$

Thus $|H| = 2|H \cap A_n|$, which means exactly half of the elements in H belong to A_n (i.e., are even). □

Problem 5.4. Let K be a proper subgroup of H , and H be a proper subgroup of G ($K \subsetneq H \subsetneq G$). If $|K| = 42$ and $|G| = 420$, what are the possible orders of H ?

Solution. By Lagrange's Theorem:

$$|K| \mid |H| \quad \text{and} \quad |H| \mid |G|.$$

Thus 42 divides $|H|$, and $|H|$ divides 420. Let $|H| = 42 \cdot m$. Then $42 \cdot m \mid 420 \implies m \mid \frac{420}{42} = 10$. The positive divisors of 10 are 1, 2, 5, 10. Since $K \subsetneq H$ is a *proper* subgroup, $|H| > |K| = 42 \implies m > 1$. Since $H \subsetneq G$ is a *proper* subgroup, $|H| < |G| = 420 \implies m < 10$. Thus $m \in \{2, 5\}$.

- If $m = 2$: $|H| = 42 \times 2 = 84$.
- If $m = 5$: $|H| = 42 \times 5 = 210$.

Therefore, the possible orders of H are 84 or 210. □

Problem 5.5. Let $H, K \leq G$. If $|H| = 12$ and $|K| = 32$, find the possible values of $|H \cap K|$.

Solution. Since $H \cap K \leq H$ and $H \cap K \leq K$, by Lagrange's Theorem:

$$|H \cap K| \mid |H| = 12 \quad \text{and} \quad |H \cap K| \mid |K| = 32.$$

Therefore, $|H \cap K|$ must divide the greatest common divisor of 12 and 32:

$$|H \cap K| \mid \gcd(12, 32) = 4.$$

The positive divisors of 4 are 1, 2, 4. Therefore, the possible values of $|H \cap K|$ are 1, 2, or 4. □

Problem 5.6. Let a and b be elements of a group G . If $|a| = 10$ and $|b| = 21$, show that $\langle a \rangle \cap \langle b \rangle = \{e\}$.

Solution. Let $x \in \langle a \rangle \cap \langle b \rangle$.

- $x \in \langle a \rangle \implies |x| \mid |a| = 10$.
- $x \in \langle b \rangle \implies |x| \mid |b| = 21$.

Thus:

$$|x| \mid \gcd(10, 21) = 1.$$

Since $|x|$ is a positive integer dividing 1, we have $|x| = 1$. The only element of order 1 in any group is the identity element e . Therefore, $x = e$, which proves:

$$\langle a \rangle \cap \langle b \rangle = \{e\}. \quad \square$$

6 The Center of a Group and Factor Group Theorems

6.1 The $G/Z(G)$ Theorem

Theorem 6.1. *Let G be a group and let $Z(G)$ denote its center:*

$$Z(G) = \{z \in G \mid zx = xz \quad \forall x \in G\}.$$

If $G/Z(G)$ is cyclic, then G is abelian (and hence $Z(G) = G$, so $|G/Z(G)| = 1$).

Proof. Assume that $G/Z(G)$ is cyclic. Then there exists some coset $aZ(G) \in G/Z(G)$ such that:

$$G/Z(G) = \langle aZ(G) \rangle.$$

Let $x, y \in G$. We must show that $xy = yx$.

Since $xZ(G)$ and $yZ(G)$ are elements of the cyclic group $G/Z(G) = \langle aZ(G) \rangle$, they must be powers of the generator $aZ(G)$. Thus, there exist integers $r, s \in \mathbb{Z}$ such that:

$$xZ(G) = (aZ(G))^r = a^r Z(G) \quad \text{and} \quad yZ(G) = (aZ(G))^s = a^s Z(G).$$

By definition of cosets, this implies:

$$x = a^r z_1 \quad \text{and} \quad y = a^s z_2 \quad \text{for some } z_1, z_2 \in Z(G).$$

Now compute the product xy :

$$\begin{aligned} xy &= (a^r z_1)(a^s z_2) \\ &= a^r (z_1 a^s) z_2 \\ &= a^r (a^s z_1) z_2 && (\text{as } z_1 \in Z(G) \implies z_1 a^s = a^s z_1) \\ &= a^{r+s} z_1 z_2. \end{aligned}$$

Next compute the product yx :

$$\begin{aligned} yx &= (a^s z_2)(a^r z_1) \\ &= a^s (z_2 a^r) z_1 \\ &= a^s (a^r z_2) z_1 && (\text{as } z_2 \in Z(G) \implies z_2 a^r = a^r z_2) \\ &= a^{s+r} z_2 z_1 \\ &= a^{r+s} z_1 z_2 && (\text{as } z_1, z_2 \in Z(G) \implies z_2 z_1 = z_1 z_2). \end{aligned}$$

Comparing both expressions:

$$xy = yx.$$

Since this holds for all $x, y \in G$, G is an abelian group. Consequently, every element commutes with all elements of G , so $Z(G) = G$. Thus $|G/Z(G)| = [G : Z(G)] = [G : G] = 1$. $\square$

6.2 Forbidden Prime Orders for $G/Z(G)$

Problem 6.1. Show that $|G/Z(G)| \neq p$, where p is any prime number.

Solution. Suppose, for contradiction, that $|G/Z(G)| = p$ for some prime p . Every group of prime order is cyclic. Therefore, $G/Z(G)$ is cyclic. By the $G/Z(G)$ theorem, if $G/Z(G)$ is cyclic, then G must be abelian. If G is abelian, then $Z(G) = G$, which implies:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{|G|}{|G|} = 1.$$

This contradicts $|G/Z(G)| = p$, since p is prime ($p \geq 2 > 1$). Therefore, $|G/Z(G)| \neq p$. □

6.3 Non-Abelian Groups of Order pq

Problem 6.2. Let G be a group of order $|G| = pq$, where p and q are prime numbers. If G is non-abelian, show that $|Z(G)| = 1$.

Solution. Since $Z(G)$ is a subgroup of G ($Z(G) \leq G$), by Lagrange's Theorem:

$$|Z(G)| \mid |G| = pq.$$

The positive divisors of pq are $1, p, q, pq$. Thus, the only possible values for $|Z(G)|$ are:

$$|Z(G)| \in \{1, p, q, pq\}.$$

We examine each case:

1. **Case 1:** $|Z(G)| = pq$. Then $|Z(G)| = |G| \implies Z(G) = G$, which means G is abelian. This contradicts the hypothesis that G is non-abelian. Thus $|Z(G)| \neq pq$.
2. **Case 2:** $|Z(G)| = p$. Then the index is:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{pq}{p} = q.$$

Since q is prime, the quotient group $G/Z(G)$ has prime order, so $G/Z(G)$ is cyclic. By the $G/Z(G)$ theorem, G must be abelian, which is a contradiction. Thus $|Z(G)| \neq p$.

3. **Case 3:** $|Z(G)| = q$. Then the index is:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{pq}{q} = p.$$

Since p is prime, $G/Z(G)$ is cyclic, which again implies G is abelian (a contradiction). Thus $|Z(G)| \neq q$.

Eliminating p, q , and pq , the only remaining possibility is:

$$|Z(G)| = 1.$$

□

6.4 Combinatorics of Cycles and Normal Subgroups of S_n

Theorem 6.2 (Formula for the Number of r -Cycles in S_n). *The number of distinct cycles of length r in the symmetric group S_n ($2 \leq r \leq n$) is:*

$$\binom{n}{r}(r-1)! = \frac{n!}{r!(n-r)!}(r-1)! = \frac{n!}{r(n-r)!} = \frac{{}^nP_r}{r}.$$

Proof. To construct an r -cycle $(a_1 a_2 \dots a_r)$:

1. Choose r distinct elements from $\{1, 2, \dots, n\}$ in $\binom{n}{r}$ ways.
2. Arrange these r elements in a circular permutation in $(r-1)!$ ways.

Multiplying yields $\binom{n}{r}(r-1)! = \frac{{}^nP_r}{r}$. □

Note (Normal Subgroups of S_n). • For all $n \neq 4$ ($n \geq 3$): the **only** normal subgroups of S_n are the trivial subgroup $\{I\}$, the alternating group A_n , and S_n itself:

$$\{I\}, \quad A_n, \quad S_n.$$

- For $n = 4$: the normal subgroups of S_4 are:

$$\{I\}, \quad V_4, \quad A_4, \quad S_4.$$

7 Commutator Subgroups (Derived Subgroups)

7.1 Definitions

Definition 7.1 (Commutator). Let G be a group and let $a, b \in G$. The element:

$$[a, b] = aba^{-1}b^{-1}$$

is called the **commutator** of the pair a and b .

Note. Observe that:

$$aba^{-1}b^{-1} = e \iff ab = ba.$$

Thus, commutators quantify the extent to which two elements fail to commute.

Definition 7.2 (Commutator / Derived Subgroup). Let S be the set of all commutators of elements of G :

$$S = \{aba^{-1}b^{-1} \mid a, b \in G\}.$$

The subgroup generated by S , denoted by $\langle S \rangle$, is called the **commutator subgroup** (or **derived subgroup**) of G . It is denoted by:

$$G' \quad \text{or} \quad G^c \quad \text{or} \quad [G, G].$$

7.2 The Five Fundamental Theorems on Commutators

Theorem 7.1. Let G be a group and G' its commutator subgroup. Then:

1. $G' = \{e\}$ if and only if G is an abelian group.
2. $G' \trianglelefteq G$.
3. G/G' is an abelian group (called the **abelianization** of G).
4. If $H \trianglelefteq G$, then G/H is abelian if and only if $G' \subseteq H$.
5. Let $H \leq G$. If $G' \subseteq H$, then $H \trianglelefteq G$.

Proof. We prove each of the five statements rigorously:

Proof of (1): $G' = \{e\} \iff G$ is abelian.

- ($\implies$) Let $G' = \{e\}$. For any $a, b \in G$, the commutator $aba^{-1}b^{-1} \in S \subseteq G' = \{e\}$.

$$\begin{aligned} aba^{-1}b^{-1} = e &\implies aba^{-1} = b \\ &\implies ab = ba. \end{aligned}$$

Thus, G is abelian.

- ($\impliedby$) Let G be abelian. Then for all $a, b \in G$:

$$ab = ba \implies aba^{-1}b^{-1} = e.$$

Thus $S = \{e\}$, so $G' = \langle S \rangle = \langle \{e\} \rangle = \{e\}$.

Proof of (2): $G' \trianglelefteq G$. Let $g \in G$ and $h \in G'$. Since every element of G' is a finite product of commutators and their inverses, it suffices to prove that the conjugate of any commutator $aba^{-1}b^{-1} \in S$ by g is again in S :

$$\begin{aligned} g(aba^{-1}b^{-1})g^{-1} &= (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gb^{-1}g^{-1}) \\ &= (gag^{-1})(gbg^{-1})(gag^{-1})^{-1}(gbg^{-1})^{-1}. \end{aligned}$$

Setting $x = gag^{-1} \in G$ and $y = gbg^{-1} \in G$, we have:

$$g(aba^{-1}b^{-1})g^{-1} = xyx^{-1}y^{-1} \in S \subseteq G'.$$

Therefore, $gG'g^{-1} \subseteq G'$ for all $g \in G$, which proves $G' \trianglelefteq G$.

Proof of (3): G/G' is an abelian group. Let $aG', bG' \in G/G'$. We must show that $(aG')(bG') = (bG')(aG')$. Notice that:

$$ab = ba(a^{-1}b^{-1}ab) = ba[a^{-1}, b^{-1}].$$

Since $[a^{-1}, b^{-1}] \in S \subseteq G'$, we have:

$$(a^{-1}b^{-1}ab)G' = G'.$$

Therefore:

$$\begin{aligned} (aG')(bG') &= abG' \\ &= ba(a^{-1}b^{-1}ab)G' \\ &= baG' \\ &= (bG')(aG'). \end{aligned}$$

Hence, G/G' is an abelian group.

Proof of (4): If $H \trianglelefteq G$, then G/H is abelian $\iff G' \subseteq H$.

- ($\implies$) Let G/H be abelian. For any $a, b \in G$:

$$\begin{aligned} (aH)(bH) &= (bH)(aH) \implies abH = baH \\ &\implies (ba)^{-1}(ab) \in H \\ &\implies a^{-1}b^{-1}ab \in H \\ &\implies [a^{-1}, b^{-1}] \in H. \end{aligned}$$

Since this holds for all elements $a, b \in G$, every commutator belongs to H . Thus $S \subseteq H$. Since H is a subgroup containing S , it must contain the subgroup generated by S :

$$\langle S \rangle = G' \subseteq H.$$

- ($\impliedby$) Conversely, let $G' \subseteq H$. For any $a, b \in G$, the commutator $aba^{-1}b^{-1} \in G' \subseteq H$.

$$\begin{aligned} aba^{-1}b^{-1} \in H &\implies ab(ba)^{-1} \in H \\ &\implies abH = baH \\ &\implies (aH)(bH) = (bH)(aH). \end{aligned}$$

Thus, G/H is abelian.

Proof of (5): Let $H \leq G$. If $G' \subseteq H$, then $H \trianglelefteq G$. Let $a \in G$ and $h \in H$. Consider the commutator of a and h :

$$aha^{-1}h^{-1} \in G'.$$

Since $G' \subseteq H$, we have $aha^{-1}h^{-1} \in H$. Now multiply by $h \in H$ on the right:

$$aha^{-1} = (aha^{-1}h^{-1})h \in H \cdot H = H.$$

Thus $aha^{-1} \in H$ for all $a \in G$ and all $h \in H$. Therefore:

$$aHa^{-1} \subseteq H \implies H \trianglelefteq G. \quad \square$$

Remark. From properties (3) and (4), we deduce that:

- G' is the **smallest** normal subgroup of G such that G/G' is abelian.
- G/G' is the **largest** abelian quotient group of G .

7.3 Explicit Computations of Commutator Subgroups

Example 7.3 (Commutator Subgroup of the Quaternion Group Q_8). Find the commutator subgroup of $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Solution. Recall that $Z(Q_8) = \{1, -1\}$. The quotient group has order:

$$[Q_8 : \{1, -1\}] = \frac{8}{2} = 4.$$

Every group of order 4 is abelian (isomorphic to either $\mathbb{Z}_4$ or V_4). Since $Q_8/\{1, -1\}$ is abelian, by Property (4) of Theorem 7.1:

$$Q'_8 \subseteq \{1, -1\}.$$

Since Q_8 is non-abelian ($ij = k \neq -k = ji$), by Property (1):

$$Q'_8 \neq \{1\}.$$

Since the only subsets of $\{1, -1\}$ are $\{1\}$ and $\{1, -1\}$, we must have:

$$\boxed{Q'_8 = \{1, -1\} = Z(Q_8)}. \quad \square$$

Example 7.4 (Commutator Subgroup of S_3). Find the commutator subgroup of the symmetric group S_3 .

Solution. $S_3 = \{I, (12), (13), (23), (123), (132)\}$. The normal subgroups of S_3 are $\{I\}, A_3, S_3$.

1. Since S_3 is non-abelian, $S'_3 \neq \{I\}$.
2. The alternating group A_3 has index 2:

$$|S_3/A_3| = \frac{6}{3} = 2.$$

Every group of order 2 is cyclic, hence abelian. Thus S_3/A_3 is abelian.

3. By Property (4):

$$S'_3 \subseteq A_3.$$

4. The order of A_3 is 3 (prime), so its only subgroups are $\{I\}$ and A_3 . Since $S'_3 \neq \{I\}$, we conclude:

$$\boxed{S'_3 = A_3}. \quad \square$$

Example 7.5 (Commutator Subgroup of A_3). Find the commutator subgroup of A_3 .

Solution. $|A_3| = 3$, which is a prime number. Every group of prime order is cyclic, and every cyclic group is abelian. Since A_3 is abelian, by Property (1):

$$\boxed{A'_3 = \{I\}}. \quad \square$$

Example 7.6 (Commutator Subgroup of S_4). Find the commutator subgroup of S_4 .

Solution. The normal subgroups of S_4 are $\{I\}, V_4, A_4, S_4$.

1. Since $|S_4/A_4| = \frac{24}{12} = 2$, S_4/A_4 is abelian. By Property (4):

$$S'_4 \subseteq A_4.$$

2. Now consider the commutator of $(1\ 2)$ and $(2\ 3)$ in S_4 :

$$\begin{aligned} [(1\ 2), (2\ 3)] &= (1\ 2)(2\ 3)(1\ 2)^{-1}(2\ 3)^{-1} \\ &= (1\ 2)(2\ 3)(1\ 2)(2\ 3) \\ &= (1\ 2\ 3)(1\ 2\ 3) \\ &= (1\ 3\ 2). \end{aligned}$$

Thus the 3-cycle $(1\ 3\ 2) \in S'_4$.

3. Since $S'_4 \leq S_4$ and contains a 3-cycle, by the Conjugacy Theorem in Section 1, S'_4 must contain all 3-cycles in S_4 .

4. The 3-cycles generate the entire alternating group A_4 . Therefore:

$$A_4 \subseteq S'_4.$$

Combining $S'_4 \subseteq A_4$ and $A_4 \subseteq S'_4$, we obtain:

$$\boxed{S'_4 = A_4}. \quad \square$$

Example 7.7 (Commutator Subgroup of A_4). Find the commutator subgroup of A_4 .

Solution. The normal subgroups of A_4 are $\{I\}, V_4, A_4$.

1. Since A_4 is non-abelian, $A'_4 \neq \{I\}$.

2. Consider the quotient by the Klein four-group V_4 :

$$|A_4/V_4| = \frac{12}{4} = 3.$$

Since 3 is prime, A_4/V_4 is cyclic, hence abelian.

3. By Property (4):

$$A'_4 \subseteq V_4.$$

4. Since $A'_4 \trianglelefteq A_4$ and $A'_4 \subseteq V_4$, A'_4 is a normal subgroup of A_4 contained in V_4 . The only normal subgroups of A_4 are $\{I\}$, V_4 , and A_4 . Since $A'_4 \neq \{I\}$ and $A'_4 \subseteq V_4$, we must have:

$$\boxed{A'_4 = V_4}.$$

□

8 Introduction to Group Actions

8.1 Definition and Axioms of a Group Action

Definition 8.1 (Group Action). Let G be a group and let A be a non-empty set. A **group action** (or **action**) of G on A is a mapping:

$$* : G \times A \rightarrow A, \quad (g, a) \mapsto g * a$$

such that the following two axioms are satisfied:

1. **Compatibility (Associativity):**

$$g_1 * (g_2 * a) = (g_1 g_2) * a \quad \forall g_1, g_2 \in G, \forall a \in A.$$

2. **Identity:**

$$e * a = a \quad \forall a \in A,$$

where e denotes the identity element of G .

Definition 8.2 (G -Set). When an action of a group G on a set A is defined, the set A is called a G -set, and we say that G **acts on** A .

8.2 Left Actions vs. Right Actions

Definition 8.3 (Left Group Action). The action defined above, where group elements operate on elements of A from the left ($* : G \times A \rightarrow A$), is referred to as an **action on A by left** or a **left action** of G on A .

Definition 8.4 (Right Group Action). We can also define an action of G on A from the right by considering a mapping:

$$* : A \times G \rightarrow A, \quad (a, g) \mapsto a * g$$

satisfying the corresponding right action axioms:

1. **Compatibility:**

$$(a * g_1) * g_2 = a * (g_1 g_2) \quad \forall g_1, g_2 \in G, \forall a \in A.$$

2. **Identity:**

$$a * e = a \quad \forall a \in A.$$